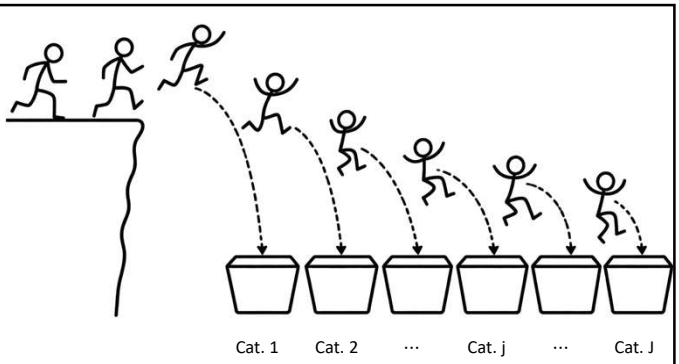
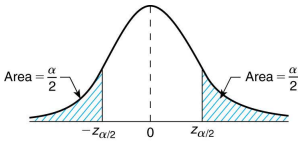
Agenda

- Earlier
  - Quantitative variables
  - Type II errors
- Today
  - Hypothesis testing for qualitative variables
  - Goodness-of-fit

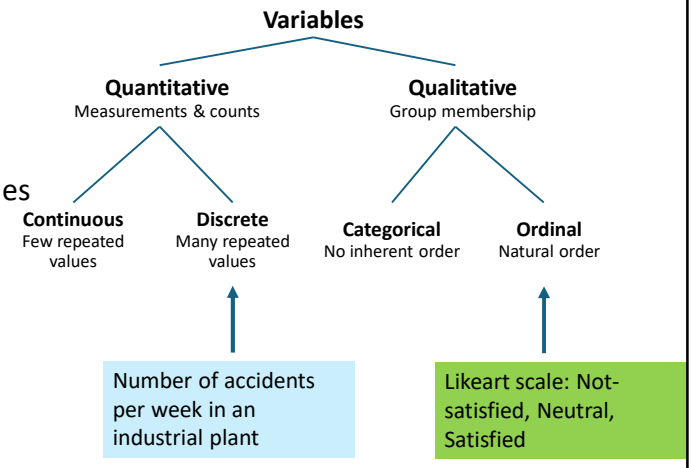




1

Types of variables

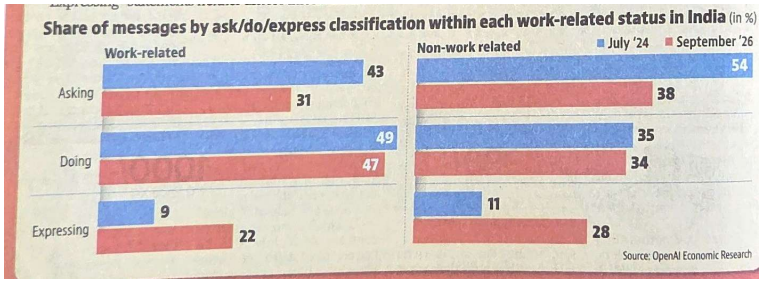
- Quantitative variables are measurements and counts
  - Variables with many repeated values are treated as discrete
  - Variables with few repeated values are treated as continuous.
- Qualitative variables describe group membership



2

Qualitative

- October 27, 2026 statistics from ChatGPT
- Blue 2024, Red 2026
  - Asking
  - Doing
  - Expressing



3

Example: Heart Attack Patients

ID	E	SYS	DIA	Blocked Arteries	STENT	TIME	Outcome	AGE	SMOKER	BETA Blockers	CHOL	SURGERY
	%	ml	ml	%	%	Months					mMol	
390	72	36	131	0	0	143	0	49	2	2	59	0
279	52	74	155	37	63	143	0	54	2	2	68	1
391	62	52	137	33	47	16	2	56	2	2	52	0
201	50	165	329	33	30	143	0	42	2	2	39	0

E: Percentage of blood in the left ventricle that is emitted (fraction)

Sys, Dia: Systolic diastolic volume

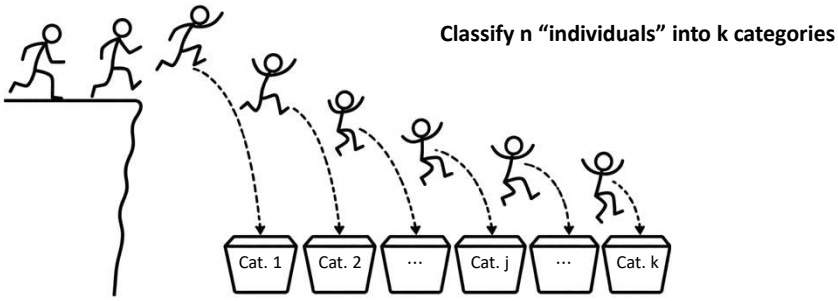
Stent: Percentage of arteries significantly lowered but not blocked

Time: Admitted until outcome (alive, sudden cardiac death, death within 30 days of heart attack …)

4

Table of counts

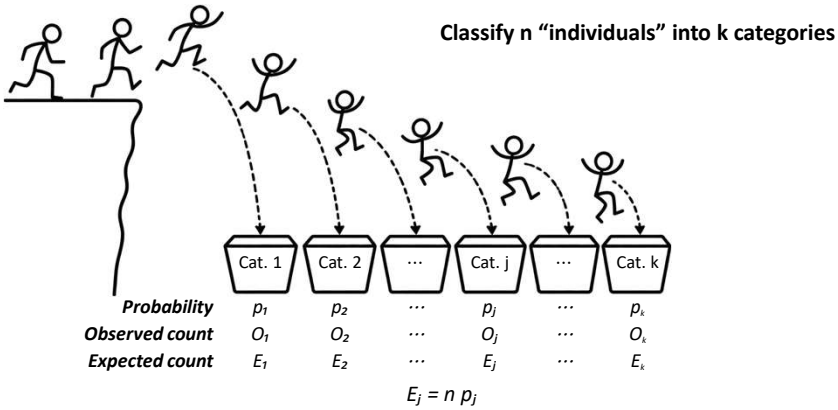
- It's usual to have a table of counts when we have one qualitative variables



5

Table of counts

- We want to test a hypothesis about the whole set of population proportions



6

Example

- Are these the rolls of a fair dice or not?
  - Fair dice: 35 for each case
- Do proportion of blood satisfy the Hardy-Weinberg model in genetics?
- Want a test statistic and thus a p-value to reject, or fail to reject our null-hypothesis

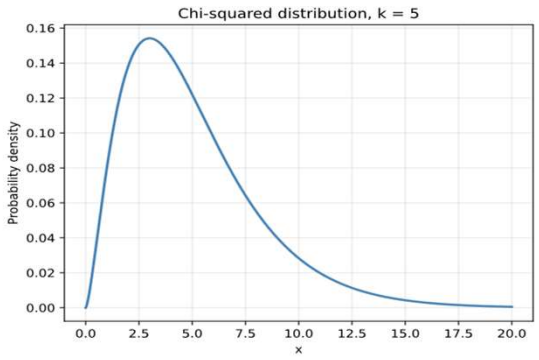
Three Blood Types				
	A	AB	B	Total
No. Observed	39	70	42	151
Proportion Observed	0.3	0.5	0.3	1.000

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
Proportion	0.124	0.190	0.176	0.124	0.205	0.181	1.000

7

Chi-square Goodness of Fit Test

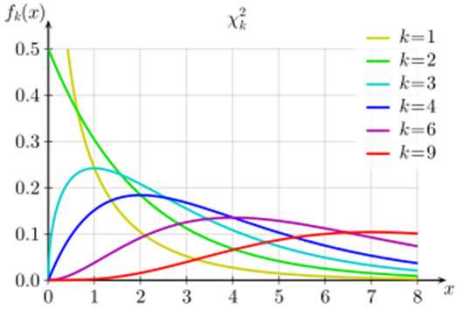
- Dice case: the test statistic is
  - $x_o^2 = \Sigma \frac{(\text{observed-expected})^2}{\text{expected}}$
  - We compute the p-value $\Pr(X^2 \geq x_0^2), X^2 \sim \chi^2(\text{df})$
  - We interpret the p-value as usual
- Degrees of freedom = k-1



8

Review: Chi-square distribution

- If $Z_1, Z_2, \ldots, Z_n$ are independent *standard* normal then
  - $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is said to have a chi-square distribution with n degrees of freedom and is denoted as $X \sim \chi_n^2$
- Example (n=1)
  - If $Z_1 \sim N(0,1)$, then $X = Z_1^2$ is χ_1^2



9

Example

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
Proportion	0.124	0.190	0.176	0.124	0.205	0.181	1.000

- Are these the rolls of a fair dice or not?
  - Null hypothesis H_0 with equal probabilities of 1/6
  - Fair dice: Count of 35 for each case
  - df = 5
  - $x_0^2 = \frac{(26 \quad)^2}{35} + \cdots + \frac{(38 \quad)^2}{35} = 7.54$
  - P-value = 0.18
- We do not have any evidence to reject the null hypothesis (i.e., we fail to reject the null hypothesis)

Intuition: Degree of freedom lost because $\Sigma p_i = 1$

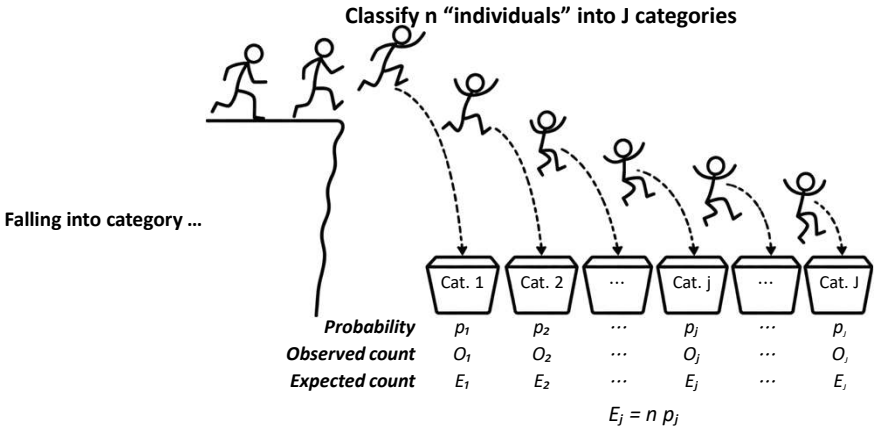
$$x_o^2 = \Sigma \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

10

Review

$$x_o^2 = \Sigma \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

- Look for observed and expected for the null hypothesis



11

Goodness of fit

- n independent variables $Y_i, \ldots, Y_n$ each taking one of the values $1,2, \ldots, k$
- Null hypothesis $H_0: p_i: i = 1, \ldots, k$ is the pmf
- Alternate hypothesis $H_a: P(Y = i) \neq p_i$ for some $i = 1, \ldots, k$
- Let $X_i, (i = 1, .., k)$ be the number of the Y_i that equal i
- Under $H_0, X_i \sim \text{Bin}(n, p_i)$
  - statistic $T = \Sigma_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$
  - When n is large, if H_o is true, then $T \sim \chi^2(k-1)$
  - Reject H_o using p-value as usual

Intuition: Don't want numerator to be large compared to the denominator

Intuition: Degree of freedom lost because $\Sigma X_i = n$

12

GoF: Goodness of fit (k=2)

- statistic $T = \Sigma_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$

How did we get to the χ^2 distribution? Consider k=2

$$
\begin{aligned}
T &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2} \\
&= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1 - p_1])^2}{n(1 - p_1)} \\
&= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1 - p_1)} \\
&= \frac{(X_1 - np_1)^2}{np_1(1 - p_1)} \qquad \text{since} \qquad \frac{1}{p} + \frac{1}{1 - p} = \frac{1}{p(1 - p)}
\end{aligned}
$$

If X_1 is Bin, then we can recognizable the denominator as the variance. For large n, we can see the standard normal appearing (it has been squared)

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